

Data Definition Language (DDL) Commands

Q. Explain the various Data Definition Language (DDL) commands in SQL. Provide the syntax and a clear example for CREATE, ALTER, DROP, TRUNCATE, and RENAME.

📌 Definition to Remember

Data Definition Language (DDL) is the subset of SQL used to define, modify, and delete the structural elements (schema) of a database — such as tables, views, and indexes. DDL commands are **auto-committed**, meaning changes are saved permanently and **cannot be rolled back**.

DDL Commands

1. CREATE

Used to create a new database or table.

```
CREATE TABLE Employee (  
    Emp_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Salary DECIMAL(10,2)  
);
```

2. ALTER

Used to modify an existing table structure — add, delete, or modify columns.

```
-- Add a new column  
ALTER TABLE Employee ADD Department VARCHAR(30);  
  
-- Modify column size  
ALTER TABLE Employee MODIFY Name VARCHAR(100);
```

3. DROP

Used to **completely delete** a table — removes structure AND all data permanently.

```
DROP TABLE Employee;
```

⚠️ *Once dropped, the table cannot be recovered without a backup.*

4. TRUNCATE

Used to **delete all rows** from a table but **keeps the table structure** intact. Faster than DELETE because it does not log individual row deletions.

```
TRUNCATE TABLE Employee;
```

5. RENAME

Used to rename an existing table.

```
RENAME TABLE Employee TO Staff;    -- MySQL
-- Oracle: ALTER TABLE Employee RENAME TO Staff;
```

Summary Table

Command	Affects	Data Lost?	Auto-Committed?
CREATE	Creates table structure	N/A	Yes
ALTER	Modifies table structure	No	Yes
DROP	Deletes table + all data	Yes (completely)	Yes
TRUNCATE	Deletes all rows	Yes (data only)	Yes
RENAME	Renames table/column	No	Yes

★ Must-Write Points (for 10 marks)

- DDL commands define the schema (structure) of a database.
- All DDL commands are **auto-committed** — changes cannot be rolled back.
- CREATE**: creates a new table/database with specified columns and constraints.
- ALTER**: modifies existing table — add/drop/modify columns without losing data.
- DROP**: permanently deletes the entire table structure AND all its data.
- TRUNCATE**: deletes all rows but preserves the table structure; faster than DELETE.
- DROP** removes structure + data; **TRUNCATE** removes data only (structure remains).

⚡ Quick Recall

DDL → Auto-committed → CREATE (make table) → ALTER (modify structure) → DROP (delete all) → TRUNCATE (delete rows, keep structure) → RENAME (change name)